$$\int \frac{1}{\sqrt{x^2+a^2}}\,dx = \ln(x+\sqrt{x^2+a^2})+C$$

令 $x = a\,\mathrm{sh}t$

$$\int \frac{1}{a\sqrt{1+(\mathrm{sh}t)^2}}\cdot a\,\mathrm{ch}t\,dt = \int 1\,dt = t+C = \mathrm{sh}^{-1}\left(\frac{x}{a}\right)+C$$

$$x = \mathrm{sh}t = \frac{e^t-e^{-t}}{2}$$

$$\Rightarrow (e^t)^2 - 2xe^t - 1 = 0$$

$$\Rightarrow e^t = \frac{2x+\sqrt{4x^2+4}}{2} = x+\sqrt{1+x^2}$$

$$\int \frac{1}{\sqrt{x^2-a^2}}\,dx = \ln\left|x+\sqrt{x^2-a^2}\right|+C$$

$$\int \mathrm{sh}x\,dx = \mathrm{ch}x+C$$

$$\int \mathrm{ch}x\,dx = \mathrm{sh}x+C$$

$\sqrt{ax^2+bx+c}$ 根式找原函数，配方，转化为 $\sqrt{a^2-x^2}$，$\sqrt{x^2+a^2}$，$\sqrt{x^2-a^2}$ 可用三角变换 $x=a\sin t$，$x=a\tan t$，$x=a\sec t$

- $$I = \int \frac{\sqrt{x^2-a^2}}{x^2}\,dx \overset{x=a\,\mathrm{ch}t}{\underset{t\geq 0}{=}} \int \frac{a\,\mathrm{sh}t}{a^2\mathrm{ch}^2t}\cdot a\cdot\mathrm{sh}t\,dt = \int \frac{\mathrm{sh}^2t}{\mathrm{ch}^2t}\,dt = 1-\frac{2}{\mathrm{ch}2t+1}\,dt$$
  $$= \int \mathrm{th}^2t\,dt$$

- $$I = \int \frac{dx}{\sqrt{x-1}\,(1+\sqrt[3]{x-1})}$$

取 $u=\sqrt[6]{x-1}$ $x=u^6+1$

$$I=\frac{6u^5du}{u^3(1+u^2)}=6\int\frac{(u^2+1)-1}{1+u^2}du=6(u-\arctan u)+C=6(\sqrt[6]{x-1}-\arctan\sqrt[6]{x-1})+C$$

欧拉变换

二次函数与直线有两个交点，一个为(0,1)，另一个则形成曲线

$$I=\int\frac{dx}{x\sqrt{1+x^2}}$$

$$\sqrt{1+x^2}=tx+1\Rightarrow x=\frac{2t}{1-t^2}$$

$$dx=\frac{2(1-t^2)+2t\cdot 2t}{(1-t^2)^2}dt=\frac{2(1+t^2)}{(1-t^2)^2}dt,\ \sqrt{1+x^2}=t\frac{2t}{1-t^2}+1=\frac{1+t^2}{1-t^2}$$

$$\int\frac{dx}{x\sqrt{1+x^2}}=\int\frac{dt}{t}=\ln|t|+C=\ln\frac{\sqrt{1+x^2}-1}{|x|}+C$$

分式积分公式 $\int\frac{du}{a^2-u^2}=\frac{1}{2a}\ln\left|\frac{a+u}{a-u}\right|+C$

分部积分

$$d(uv)=udv+vdu\Rightarrow\int udv=uv-\int vdu$$

$$I=\int x\sin x\,dx=-\int x\,d\cos x=-x\cos x+\int\cos x\,dx$$

$$I=\int xe^xdx=\int x\,de^x=xe^x-e^x+C$$

$$I=\int x\ln x\,dx=\int\ln x\,d(\tfrac{1}{2}x^2)=\tfrac{1}{2}x^2\ln x-\tfrac{1}{2}\int x^2d\ln x=\tfrac{1}{2}x^2\ln x-\tfrac{1}{2}\int x\,dx=\tfrac{1}{2}x^2\ln x-\tfrac{1}{4}x^2+C$$

$$I=\int x\arctan x\,dx$$

$$I_k=\int\frac{dx}{(x^2+a^2)^k}=\frac{x}{(x^2+a^2)^k}-\int x(-k)(x^2+a^2)^{-k-1}\cdot 2x\,dx$$

$$=\frac{x}{(x^2+a^2)^k}+2k\underbrace{\int\frac{dx}{(x^2+a^2)^k}}_{I_k}-2ka^2\underbrace{\int\frac{1}{(x^2+a^2)^{k+1}}dx}_{I_{k+1}}$$

递推. $\Rightarrow I_{k+1}=\frac{1}{2ka^2}\left(\frac{x}{(x^2+a^2)^k}+(2k-1)I_k\right)$

$$I_2=\frac{1}{2a^2}\left(\frac{x}{a^2+x^2}+I_1\right)=\frac{1}{2a^2}\left(\frac{x}{a^2+x^2}+\frac{1}{a}\arctan\frac{x}{a}\right)+C$$

分部积分适用于①幂函数与超越函数dx　$\int x\ln x\,dx$　$\int x\sin x\,dx$

选择幂函数或导数简单者为u

② 对数函数 / 反三角函数单独积分.

$\int\arctan x\,dx,\ u=\arctan x$　　$du=\frac{1}{1+x^2}dx$　　$dv=dx$　$v=x$

$$=x\arctan x-\int x\cdot\frac{1}{1+x^2}dx=x\arctan x-\frac{1}{2}\int\frac{d(1+x^2)}{1+x^2}$$

$$=x\arctan x-\frac{1}{2}\ln(1+x^2)+C$$

③ 循环型积分　$\int e^{ax}\sin bx\,dx$　$\int e^{ax}\cos bx\,dx$　(两次分部)

④ 分式与超越函数结合　$\int\frac{\ln x}{x^2}dx$

反对幂指三

u — dv

三角函数有理式积分

$\sin x = \dfrac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}$

$\cos x = \dfrac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}$

$\tan x = \dfrac{2\tan\frac{x}{2}}{1-\tan^2\frac{x}{2}}$

① $\int R(\sin x)\cos dx \overset{t=\sin x}{=} \int R(t)\,dt$

② $\int R(\cos x)\sin dx \overset{t=\cos x}{=} -\int R(t)\,dt$

③ $\int R(\tan x)\,dx \underset{x\in(-\frac{\pi}{2},\frac{\pi}{2})}{\overset{t=\tan x}{=}} \int R(t)\,dt$

若加上"−" 有 $\int R(-\sin x)\cos dx = -\int R(\sin x)\cos dx$

则可提出 $\sin x$，剩下仅有 $\sin^2 x\ (1-\cos^2 x)$ 项，转化为②

加上"−"号后变号，第①种或第②种

加上"−"号后不变号，给另一种加上"−" 第③种。 齐次.

其它情况，例如 $\int \dfrac{1}{\sin x+\cos x}dx$

$\int \tan^2 x\sec x\,dx$

$\int \dfrac{3\sin x+2\cos x}{2\sin x+3\cos x}dx$

例 $I = \int \dfrac{1}{1-2a\cos x+a^2}dx. \quad (|a|<1,\ |x|<\pi)$

$t = \tan\frac{x}{2}$

$x = 2\arctan t$

$t \in (-\infty,+\infty)$

$\Rightarrow I = \int \dfrac{\frac{2dt}{1+t^2}}{1-2a\frac{1-t^2}{1+t^2}+a^2} = \int \dfrac{2dt}{(1+a)^2t^2+(1-a)^2} = \int \dfrac{\frac{2(1-a)}{1+a}\,d\left(\frac{1+a}{1-a}t\right)}{(1-a)^2\left[1+\left(\frac{1+a}{1-a}t\right)^2\right]} = \dfrac{2}{1-a^2}\arctan\left(\dfrac{1+a}{1-a}\tan\dfrac{x}{2}\right)+C$

例 $I = \int \dfrac{dx}{4+5\sin^2 x} \overset{t=\tan\frac{x}{2}}{=} \int \dfrac{\frac{2}{1+t^2}dt}{4+5\cdot\left(\frac{2t}{1+t^2}\right)^2} = \int \dfrac{2(1+t^2)}{4(1+t^2)^2+20t^2}dt$ 计算量大！

$\underset{x=\arctan t}{\overset{t=\tan x}{=}} \int \dfrac{\frac{dt}{1+t^2}}{4+5\frac{t^2}{1+t^2}} = \int \dfrac{dt}{4+4t^2+5t^2} = \dfrac{1}{4}\int \dfrac{\frac{2}{3}d\frac{3}{2}t}{1+\left(\frac{3}{2}t\right)^2} = \dfrac{1}{6}\arctan\dfrac{3t}{2}+C = \dfrac{1}{6}\arctan\dfrac{3\tan x}{2}+C$

但仅在 $x\in(-\frac{\pi}{2},\frac{\pi}{2})$ 时成立．好在 $\sin^2 x$ 周期为π.
作周期延拓即可

例 $I=\int \frac{1}{1+\tan x}dx$

$$\overset{t=\tan x}{=}\int \frac{\frac{dt}{1+t^2}}{1+t} = \int \left(\frac{A}{1+t}+\frac{Bt+C}{1+t^2}\right)dt = \frac{1}{2}\int\left(\frac{1}{1+t}+\frac{1-t}{1+t^2}\right)dt = \frac{1}{2}(\ln|1+t|+\arctan t)-\frac{1}{2}\ln(1+t^2)+C$$

$$=\frac{1}{2}x+\frac{1}{2}\ln|\cos x+\sin x|+C$$

根式积分

$$I=\int R\left(x,\sqrt[n]{\frac{ax+b}{\alpha x+\beta}}\right)dx$$

$t=\sqrt[n]{\frac{ax+b}{\alpha x+\beta}}$ 则 $t^n=\frac{ax+b}{\alpha x+\beta}$, $x=\frac{\beta t^n-\beta}{a-\alpha t^n}$

$I=\int \frac{1}{x}\sqrt{\frac{1+x}{x}}\,dx$ 令 $t=\sqrt{\frac{1+x}{x}}$ 则 $x=\frac{1}{t^2-1}$, $dx=\frac{-2t}{(t^2-1)^2}dt$

$t^2=\frac{1+x}{x}\Rightarrow xt^2=1+x\Rightarrow x=\frac{1}{t^2-1}$

$$=-2\int\frac{t^2-1+1}{t^2-1}dt=-2\int\left[1+\frac{1}{2}\left(\frac{1}{t-1}-\frac{1}{t+1}\right)\right]dt$$

$$=-2\left(t+\frac{1}{2}\ln\left|\frac{t-1}{t+1}\right|+C\right)=-2\sqrt{\frac{1+x}{x}}+\ln\frac{\sqrt{\frac{1+x}{x}}+1}{\left|\sqrt{\frac{1+x}{x}}-1\right|}+C$$

3.1.5 常见的基本积分公式

(1) $\int x^{\alpha} \mathrm{d}x = \frac{1}{\alpha+1}x^{\alpha+1} + C(\alpha \neq -1)$

(2) $\int \frac{1}{x}\,\mathrm{d}x = \ln|x| + C$

(3) $\int a^x\,\mathrm{d}x = \frac{a^x}{\ln a} + C(a>0, a\neq 1)$

(4) $\int \mathrm{e}^x\,\mathrm{d}x = \mathrm{e}^x + C$

(5) $\int \sin x\,\mathrm{d}x = -\cos x + C$

(6) $\int \cos x\,\mathrm{d}x = \sin x + C$

(7) $\int \sec^2 x dx = \tan x + C$

(8) $\int \csc^2 x dx = -\cot x + C$

(9) $\int \sec x \tan x dx = \sec x + C$

(10) $\int \csc x \cot x\,\mathrm{d}x = -\csc x + C$

(11) $\int \sec x dx = \ln|\sec x + \tan x| + C$

(12) $\int \csc x\,\mathrm{d}x = -\ln|\csc x + \cot x| + C = \ln|\csc x - \cot x| + C$

(11) $= \int \frac{\cos x}{1-\sin^2 x}\,dx$

$= \frac{1}{2}\int \left(\frac{1}{1+u} + \frac{1}{1-u}\right) du$

$= \frac{1}{2}\ln\left|\frac{1+\sin x}{1-\sin x}\right| + C$

$\frac{(1+\sin x)^2}{1-\sin^2 x}$

(13) $\int \frac{\mathrm{d}x}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a} + C$

(14) $\int \frac{\mathrm{d}x}{a^2-x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C$

(15) $\int \frac{\mathrm{d}x}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a} + C$

(16) $\int \frac{\mathrm{d}x}{\sqrt{x^2+a^2}} = \ln\left|x+\sqrt{x^2+a^2}\right| + C$

(17) $\int \frac{\mathrm{d}x}{\sqrt{x^2-a^2}} = \ln\left|x+\sqrt{x^2-a^2}\right| + C$

笔记 背诵方法：同导数一样，有关 $\tan x, \sec x$ 为正；有关 $\cot x, \csc x$ 为负

$e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + C$

$e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx) + C$

$\int\sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} + C$

$\int\sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\ln(x+\sqrt{x^2+a^2}) + C$

$\int\sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln|x+\sqrt{x^2-a^2}| + C$